

30231053-0: Electromagnetic Fields and Waves

Tsinghua University, Spring Semester 2025

Fridays, 9:50 am – 12:15 am, 6A201

Tencent (VooV) Meeting ID: 400-3326-6709

Instructor

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Teaching Assistants

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Office Hours

Thursdays, 2:00 to 4:00 pm (appointments needed).

Course Description

Electromagnetism is fundamental to a wide range of physical phenomena and engineering applications, from wireless communication and radar systems to optical imaging and fiber-optics. This course provides a comprehensive introduction to the theory and applications of electromagnetic fields and waves, covering key concepts such as Maxwell's equations, wave propagation, waveguides, and radiation. Emphasis will be placed on both analytical techniques and physical intuition, enabling students to develop a deep understanding of how electromagnetic fields and waves behave in different media and how they can be harnessed in modern technology. Through theoretical analysis and practical examples, this course aims to build a strong foundation for students pursuing further studies or careers in electromagnetics, optics, and related fields.

Prerequisites/Recommendations

Prior knowledge of college-level math and physics is a prerequisite.

Goals

This course provides a rigorous and systematic exploration of electromagnetic theory, emphasizing both mathematical precision and physical intuition. Through a structured and logical progression of concepts, students will gain a deep appreciation for the elegance and coherence of electromagnetism's mathematical framework. The course aims to equip students with the analytical skills necessary to tackle complex electromagnetic problems while fostering a strong conceptual foundation. Beyond technical proficiency, we seek to cultivate independent thinking, critical reasoning, and the ability to apply electromagnetic principles to diverse scientific and engineering challenges. By the end of the course, students will have developed not only a comprehensive understanding of electromagnetism but also the intellectual agility to approach advanced topics.

Textbooks

1. Main textbook: *Introduction to Electrodynamics*, David J. Griffiths, Cambridge University Press, Fourth Edition (ISBN: 978-1108420419).
(Electronic version downloadable at <http://reserves.lib.tsinghua.edu.cn/Home>).
2. Reference textbook: *The Feynman Lectures on Physics, Volume II*
(Electronic version downloadable at https://www.feynmanlectures.caltech.edu/II_toc.html)
3. Reference textbook (in Chinese): 《电动力学》郭硕鸿 著，高等教育出版社，第三版 (ISBN:978-7040239249).

Web Resources

1. The following link provides an online tutorial with many figures/videos, which will help you better visualize the EM fields:
<http://web.mit.edu/8.02t/www/802TEAL3D/visualizations/guidedtour/Tour.htm>
2. Prof. Walter Lewin's lecture series features interesting experimental demonstrations and is available on YouTube and Bilibili.

Grading

Homework 20%; midterm exam 40%; final exam 40%.

Schedule

Week	Date	Topics
1	Feb 21	Basics of vector analysis.
2	Feb 28	Electrostatics, Coulomb's law, Gauss's law, conductors.
3	Mar 7	Electric potential, dielectric and polarization.
4	Mar 14	Electric boundary conditions, uniqueness theorem, method of images, separation of variables.
5	Mar 21	Magnetostatic fields, steady currents.
6	Mar 28	Properties of magnetostatic fields, vector potential, magnetic fields in matter, magnetic boundary conditions.
8	Apr 11	Force and energy of electric and magnetic dipoles, A-B effect, electromagnetic induction, Faraday's law.
9	Apr 18	Displacement current, Maxwell's equations.
10	Apr 25	Mid-term exam (open-book, 2 hours).
11	Apr 29	Poynting's theorem, EM waves, plane waves.
12	May 9	General solutions to the wave equations, dipole radiation, antenna.
13	May 16	Linear light-matter interaction, the origin of refractive index, absorption, dispersion.
14	May 23	Reflection and transmission of plane waves at an interface.
15	May 30	Waveguides, diffraction.
16	Jun 6	TBD: Course review or no class, depending on the final exam date
17	TBD	Final exam (open-book, 2 hours).